Assessment Task 2: Project

Project Brief
Read thought the brief below take note of the needs and requirements outlined in the brief. (The learners do not need to complete any tasks for Part 1. The learners simple need read and understand the requirements of the brief.)

Introduction
Since its inception, Cyber Consulting has been defining the home automation industry. Founded in 2011, Cyber Consulting has been led by folks who've spent most of their careers in startups and combine serious technical knowledge with an inspiring vision of what the future of cyber security could be. They bring together a passionate, dynamic team of game changers.

Problem
Cyber Consulting are looking for talented programmers to join their team. They are looking for developers who can work as part of a team to create cyber security python solutions. They are interested in teams working with scripts to automate progresses in cyber security and showcase innovations.

Project requirements
You need to research cyber security python solutions and develop scripts that contribute to the creation interesting cyber security python projects (example authentication page, game, password reset,etc.). You must create at least two (2) scripts that can showcase your talent.

General requirements
- Projects must be written in the Python programming language
- The you must the Spyder, relp.it or Anaconda IDE to create the project
- Pseudo code is used to show the script idea
- The pseudo code is checked by at least 1 peer for missing logic and errors
- The script works without errors
-

Technical requirements
- Python
- This custom code must be documented. (See Part 4 Question 5)
- Code editor such as Visual Studio Code or repl.it

Part 1 - Identifying the project requirements
Based on the information you have gathered from the brief complete the following:

1. After reading and interpreting the brief, what is the Integrated Development Environment (IDE) that will be used?	*Replit and PyCharm* *Link for the code:* *https://replit.com/join/qogldxlric-osasson*
2. Identify the programming language to be used in the projects.	*Python*
3. List 3 or 4 projects that your team may work on building.	*Signup system* *Password generator* *Log in system*
4. Identify the protocols and object models that will be used in the projects.	*Re* *Random*

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	String *Csv* *Colorama* *Pandas*

Approval and Feedback
Once you have identified the project requirements obtain approval from the client's representative.

Before moving forward learners will need to organize a meeting with their facilitator to obtain approval and feedback before moving on. The following check list is to be complete by the facilitator.

Checklist (To be completed by the learner's facilitator)	**Yes**	**No**
1. The learner has demonstrated a clear understanding of the client's business and core project needs.	☐	☐
2. The learner has a clear understanding of the IDE and technical requirements.	☐	☐
3. The learner has investigate the programming language that will be used..	☐	☐
4. The learner has determined at least three (3) projects and documentation required.	☐	☐
5. The learner has determined the protocols and object models that will be used in the projects	☐	☐

Assessor Name	*Insert Name*	**Assessor Signature**	*Insert Signature*	**Date**	*Insert Date*

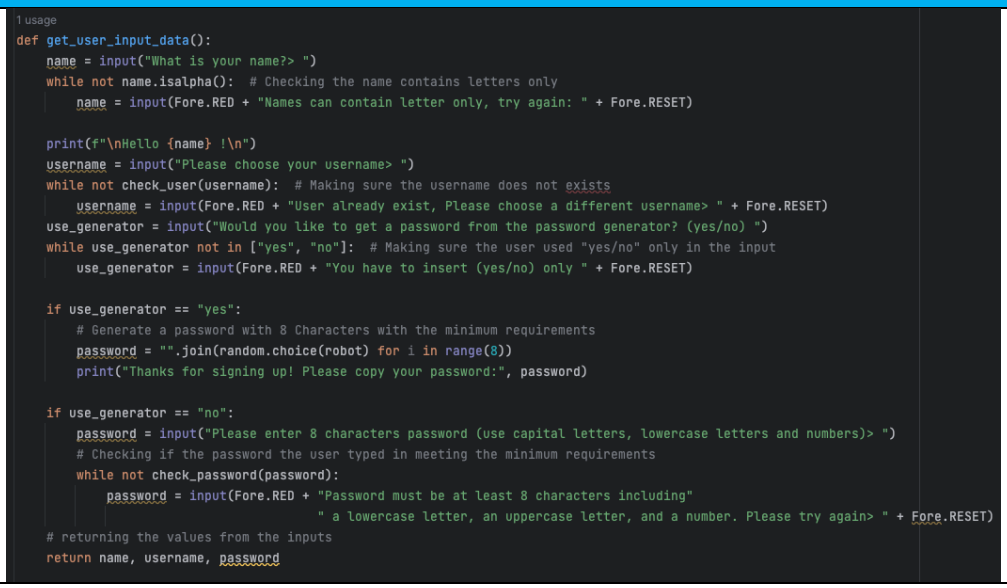
Part 2 - Design and scripts for Cyber security python portfolio
It's time to start building. Create pseudo code for a Cyber security python solution

1. Show examples of created pseudo code for all the five project scripts	Get user input data function: Taking the user input as: name, username WHILE name NOT with letters only User cannot proceed WHILE username in 'users' file PRINT "User exists" WHILE username NOT in 'users' file Ask the user if he would like to get a password from the generator IF "yes" Generate a password and show it to the user IF "no" Ask user to choose a password which meets the minimum requirements WHILE NOT meeting the minimum requirements

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	Tell the user it has to meet the minimum requirements END IF END WHILE RETURN name, username, password
2. Review of pseudo code by at least 1 peer. Feedback should include missing logic and errors	*GM1 script reviewed by GM2: Comments - The above code won't work as you have not indented the Print statement.* *GM2 script reviewed by GM3: - Comments*

Part 3 - Design and Develop Cyber security python project
Develop pseudocode for Cyber security python solution

1. Screenshot one script you have translated from pseudo code	

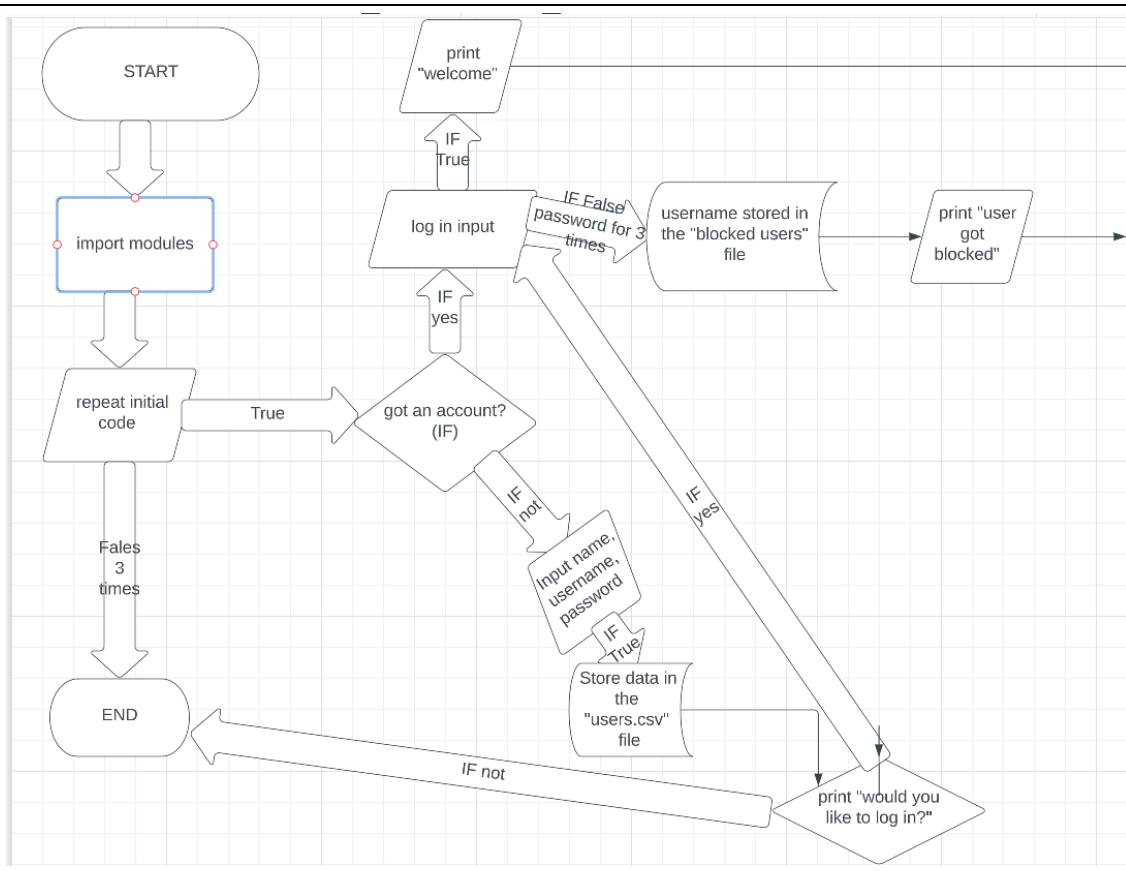
```
1 usage
def get_user_input_data():
    name = input("What is your name?> ")
    while not name.isalpha():  # Checking the name contains letters only
        name = input(Fore.RED + "Names can contain letter only, try again: " + Fore.RESET)

    print(f"\nHello {name} !\n")
    username = input("Please choose your username> ")
    while not check_user(username):  # Making sure the username does not exists
        username = input(Fore.RED + "User already exist, Please choose a different username> " + Fore.RESET)
    use_generator = input("Would you like to get a password from the password generator? (yes/no) ")
    while use_generator not in ["yes", "no"]:  # Making sure the user used "yes/no" only in the input
        use_generator = input(Fore.RED + "You have to insert (yes/no) only " + Fore.RESET)

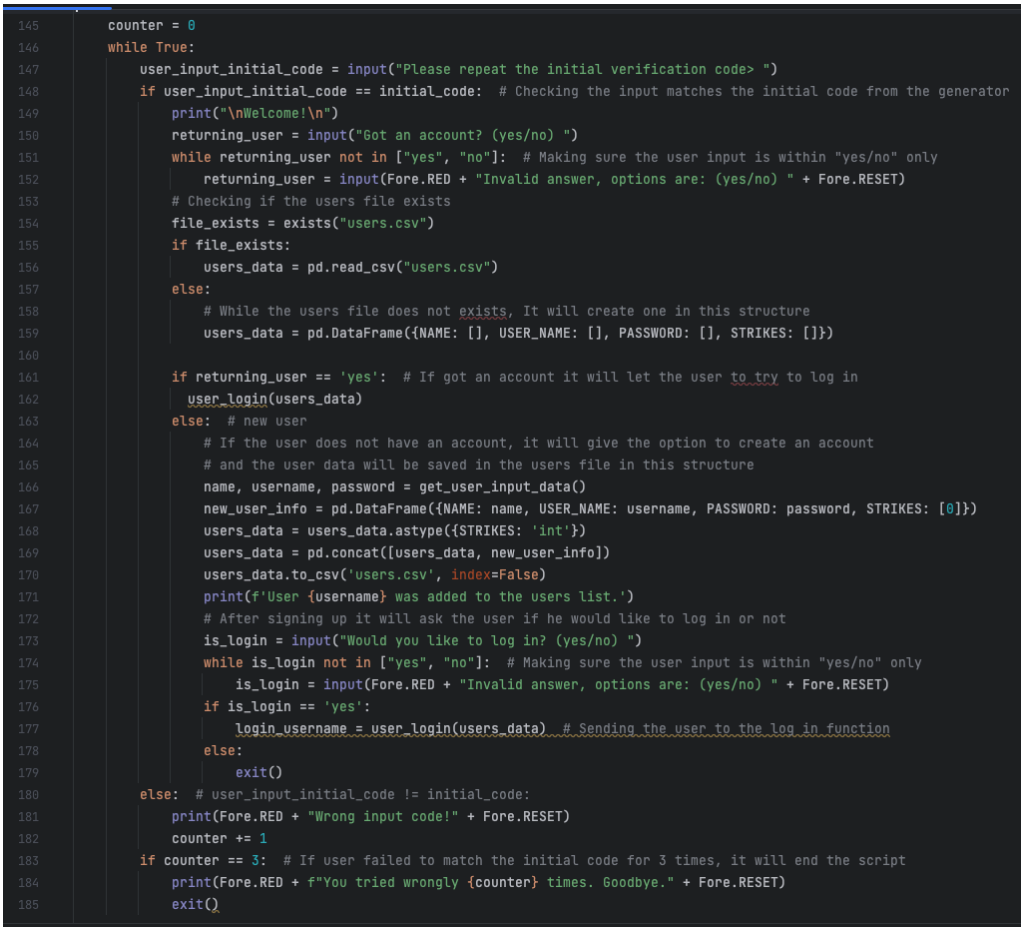
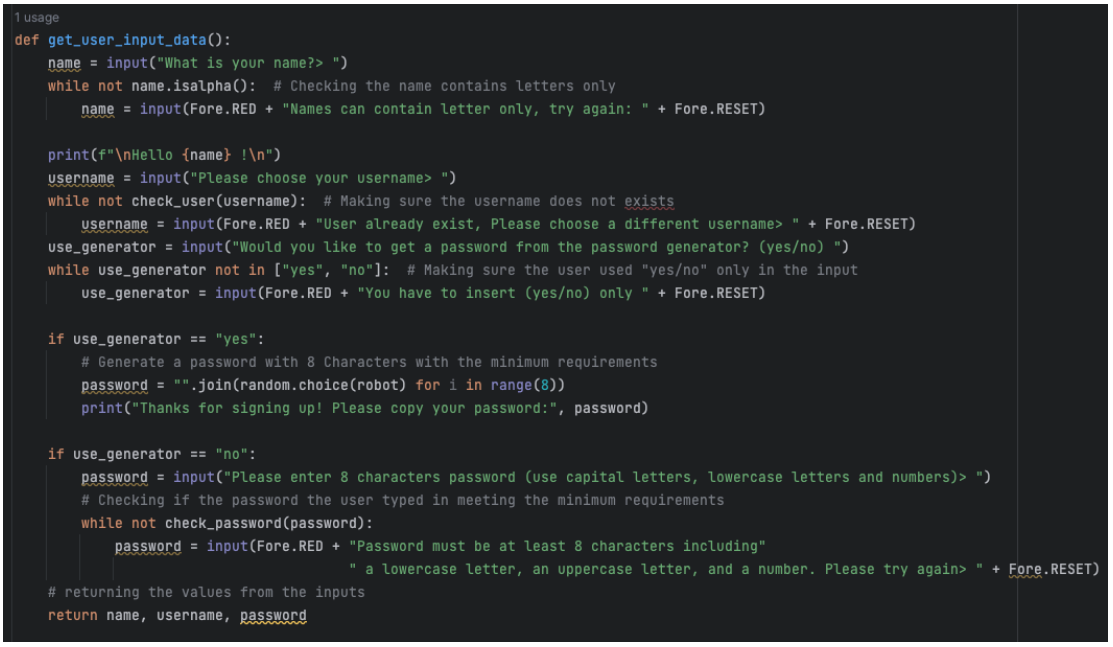
    if use_generator == "yes":
        # Generate a password with 8 Characters with the minimum requirements
        password = "".join(random.choice(robot) for i in range(8))
        print("Thanks for signing up! Please copy your password:", password)

    if use_generator == "no":
        password = input("Please enter 8 characters password (use capital letters, lowercase letters and numbers)> ")
        # Checking if the password the user typed in meeting the minimum requirements
        while not check_password(password):
            password = input(Fore.RED + "Password must be at least 8 characters including"
                                        " a lowercase letter, an uppercase letter, and a number. Please try again> " + Fore.RESET)
    # returning the values from the inputs
    return name, username, password
```

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2. Screenshot algorithm(flowchart) for all five scripts that has been tested to work in all situations with automation functionality	

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3. Screenshot one example of how your code has a guaranteed end	
4. Screenshot one example of your code with structure, sequence, selection and iteration	

```
    counter = 0
    while True:
        user_input_initial_code = input("Please repeat the initial verification code> ")
        if user_input_initial_code == initial_code:  # Checking the input matches the initial code from the generator
            print("\nWelcome!\n")
            returning_user = input("Got an account? (yes/no) ")
            while returning_user not in ["yes", "no"]:  # Making sure the user input is within "yes/no" only
                returning_user = input(Fore.RED + "Invalid answer, options are: (yes/no) " + Fore.RESET)
            # Checking if the users file exists
            file_exists = exists("users.csv")
            if file_exists:
                users_data = pd.read_csv("users.csv")
            else:
                # While the users file does not exists, It will create one in this structure
                users_data = pd.DataFrame({NAME: [], USER_NAME: [], PASSWORD: [], STRIKES: []})

            if returning_user == 'yes':  # If got an account it will let the user to try to log in
              user_login(users_data)
            else:  # new user
                # If the user does not have an account, it will give the option to create an account
                # and the user data will be saved in the users file in this structure
                name, username, password = get_user_input_data()
                new_user_info = pd.DataFrame({NAME: name, USER_NAME: username, PASSWORD: password, STRIKES: [0]})
                users_data = users_data.astype({STRIKES: 'int'})
                users_data = pd.concat([users_data, new_user_info])
                users_data.to_csv('users.csv', index=False)
                print(f'User {username} was added to the users list.')
                # After signing up it will ask the user if he would like to log in or not
                is_login = input("Would you like to log in? (yes/no) ")
                while is_login not in ["yes", "no"]:  # Making sure the user input is within "yes/no" only
                    is_login = input(Fore.RED + "Invalid answer, options are: (yes/no) " + Fore.RESET)
                if is_login == 'yes':
                    login_username = user_login(users_data)  # Sending the user to the log in function
                else:
                    exit()
        else:  # user_input_initial_code != initial_code:
            print(Fore.RED + "Wrong input code!" + Fore.RESET)
            counter += 1
        if counter == 3:  # If user failed to match the initial code for 3 times, it will end the script
            print(Fore.RED + f"You tried wrongly {counter} times. Goodbye." + Fore.RESET)
            exit()
```

```
1 usage
def get_user_input_data():
    name = input("What is your name?> ")
    while not name.isalpha():  # Checking the name contains letters only
        name = input(Fore.RED + "Names can contain letter only, try again: " + Fore.RESET)

    print(f"\nHello {name} !\n")
    username = input("Please choose your username> ")
    while not check_user(username):  # Making sure the username does not exists
        username = input(Fore.RED + "User already exist, Please choose a different username> " + Fore.RESET)
    use_generator = input("Would you like to get a password from the password generator? (yes/no) ")
    while use_generator not in ["yes", "no"]:  # Making sure the user used "yes/no" only in the input
        use_generator = input(Fore.RED + "You have to insert (yes/no) only " + Fore.RESET)

    if use_generator == "yes":
        # Generate a password with 8 Characters with the minimum requirements
        password = "".join(random.choice(robot) for i in range(8))
        print("Thanks for signing up! Please copy your password:", password)

    if use_generator == "no":
        password = input("Please enter 8 characters password (use capital letters, lowercase letters and numbers)> ")
        # Checking if the password the user typed in meeting the minimum requirements
        while not check_password(password):
            password = input(Fore.RED + "Password must be at least 8 characters including"
                                        " a lowercase letter, an uppercase letter, and a number. Please try again> " + Fore.RESET)
    # returning the values from the inputs
    return name, username, password
```

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5. Screenshot one script that manipulates variables

```
# If user exists, it will ask for the password
strikes_counter = 0
while strikes_counter < 3:
    password = input("Please enter your password> ")
    while login_password_check(users_data, username, password):  # Checking that the user matches its password
        print(f"Welcome {username}")
        exit()

    else:  # password incorrect
        print(Fore.RED + f"Password did not match username {username}! Please try again" + Fore.RESET)
        strikes_counter += 1

if strikes_counter == 3:
    # If user failed to match the password 3 times in a raw, it will be added to the blocked users file
    with open('blocked_users.csv', 'a', newline='') as csvfile:
        fieldnames = ({username})
        writer = csv.DictWriter(csvfile, fieldnames=fieldnames)
        writer.writeheader()
        print("\033[1;31;40m" "You tried wrongly 3 times. Your account is blocked."
              " Please contact the administrator.")
        exit()
```

Part 4 – Documentation and Debugging

1. Screenshot of debugging syntax and semantic errors

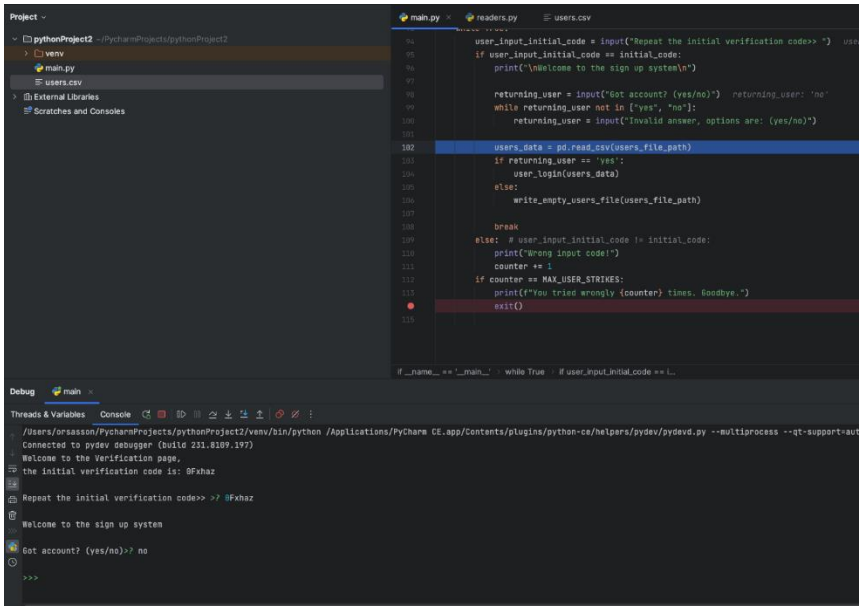
```
/Users/orsasson/PycharmProjects/pythonProject2/venv/bin/python /Users/orsasson/PycharmProjects/pythonProject
Welcome to the Verification page,
the initial verification code is: 2Nn90Q

Repeat the initial verification code>> 2Nn90Q

Welcome to the sign up system

Got account? (yes/no)no
Traceback (most recent call last):
  File "/Users/orsasson/PycharmProjects/pythonProject2/main.py", line 102, in <module>
    users_data = pd.read_csv(users_file_path)
  File "/Users/orsasson/PycharmProjects/pythonProject2/venv/lib/python3.9/site-packages/pandas/util/_decorat
    return func(*args, **kwargs)
```

It did not use the file properly



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```
1 usage
def write_empty_users_file(users_data, users_file_path):
    users_data = pd.DataFrame({NAME: [], 'User name': [], 'Password': [], 'Strikes': []})
    users_data.to_csv(users_file_path, index=False)
    return


def add_user_to_users_file(users_data, users_file_path):
    input_name, input_username, input_password = get_user_input_data(robot)
    users_data = pd.concat([users_data, pd.DataFrame({NAME: input_name, 'User name': input_username,
                                                      'Password': input_password, 'Strikes': [0]})])
    users_data.to_csv(users_file_path, index=False)
    return
```

```
users_data = pd.read_csv(users_file_path)
if returning_user == 'yes':
    user_login(users_data)
else:
    write_empty_users_file(users_file_path)
```

I have changed the structure of the code and the way It reads the file or creates a new one

```
returning_user = input("Got an account? (yes/no) ")
while returning_user not in ["yes", "no"]:  # Making sure the user input is within "yes/no" only
    returning_user = input(Fore.RED + "Invalid answer, options are: (yes/no) " + Fore.RESET)
# Checking if the users file exists
file_exists = exists("users.csv")
if file_exists:
    users_data = pd.read_csv("users.csv")
else:
    # While the users file does not exists, It will create one in this structure
    users_data = pd.DataFrame({NAME: [], USER_NAME: [], PASSWORD: [], STRIKES: []})

if returning_user == 'yes':  # If got an account it will let the user to try to log in
  user_login(users_data)

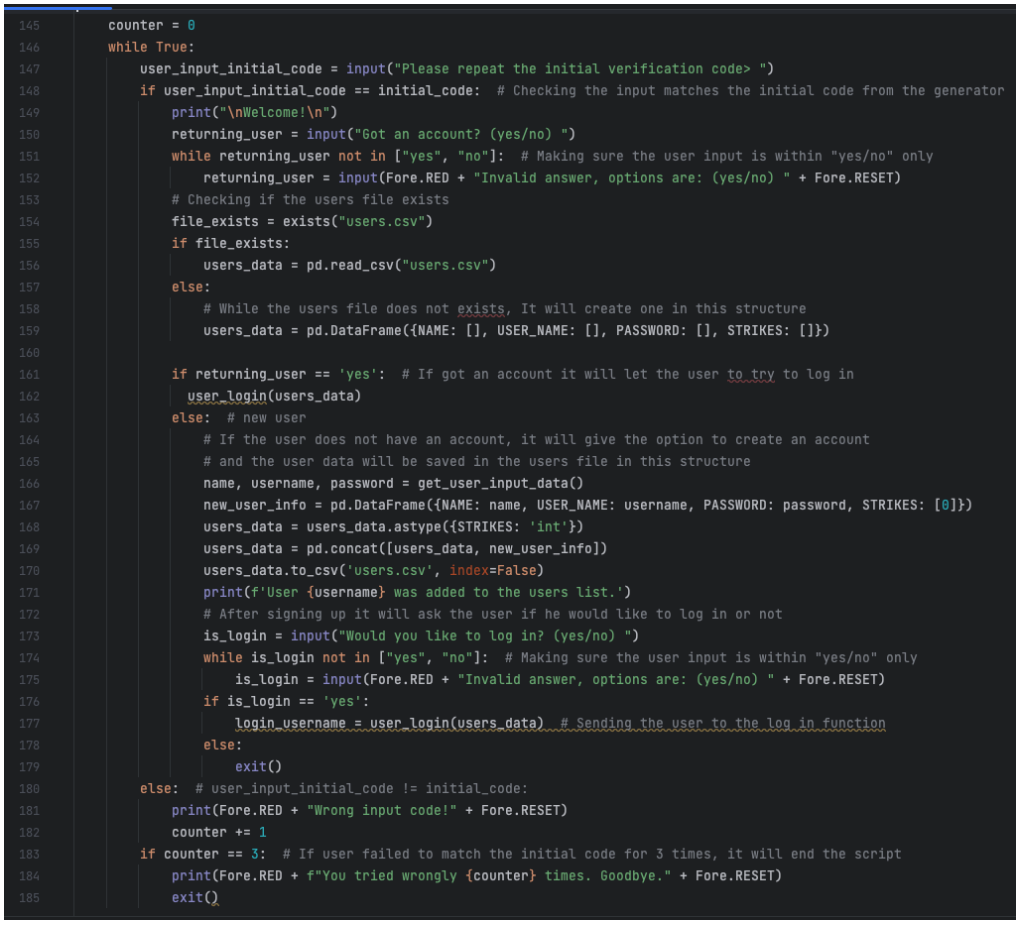

else:  # new user
    # If the user does not have an account, it will give the option to create an account
    # and the user data will be saved in the users file in this structure
    name, username, password = get_user_input_data()
    new_user_info = pd.DataFrame({NAME: name, USER_NAME: username, PASSWORD: password, STRIKES: [0]})
    users_data = users_data.astype({STRIKES: 'int'})
    users_data = pd.concat([users_data, new_user_info])
    users_data.to_csv('users.csv', index=False)
    print(f'User {username} was added to the users list.')
    # After signing up it will ask the user if he would like to log in or not
    is_login = input("Would you like to log in? (yes/no) ")
    while is_login not in ["yes", "no"]:  # Making sure the user input is within "yes/no" only
        is_login = input(Fore.RED + "Invalid answer, options are: (yes/no) " + Fore.RESET)
    if is_login == 'yes':
        login_username = user_login(users_data)  # Sending the user to the log in function
    else:
        exit()
```

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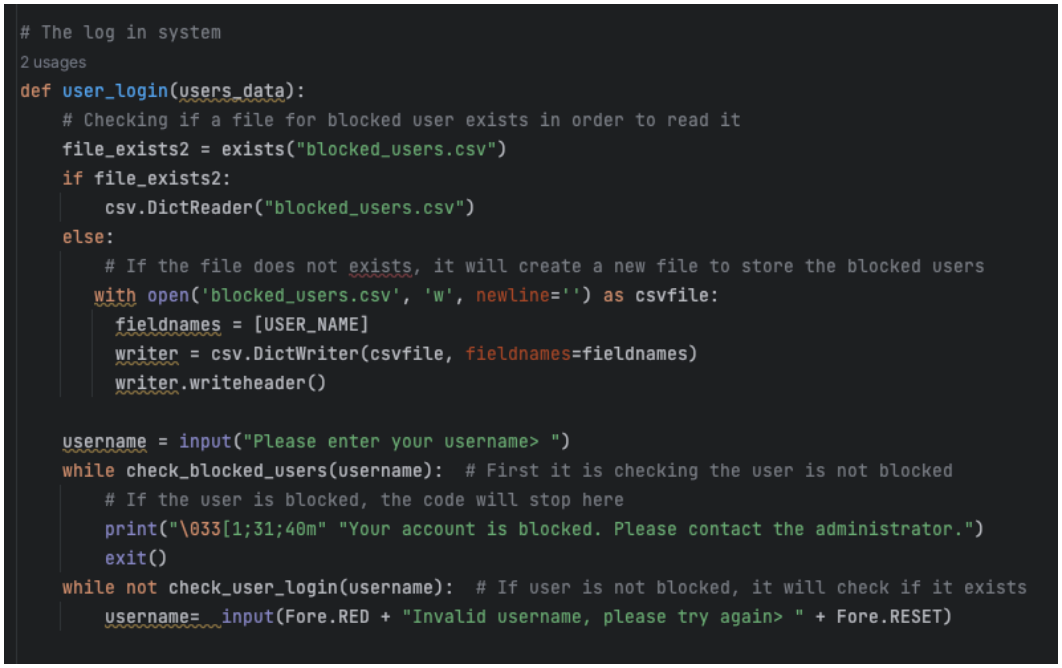
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2. Write up internal technical-level documentation of the scripts you have created	*I imported the following modules:* 1. *Re* 2. *Random* 3. *String* 4. *Csv* 5. *Colorama* 6. *pandas* *In the script I wrote there are 7 functions apart of the main script, which is functioning when gets call the main script along the way.* 1. *Check_password: Making sure the password the user type in when signing up, is meeting th minimum requirements.* 2. *Check_user: checking if the username exists when signing up, so every user has a unique username.* 3. *Check_blocked_users: checking if a user on the blocked users list when trying to log in.* 4. *Check_user_login: checking if the user on the users file, so it can let it in.* 5. *Login_password_check: making sure the username matches the password.* 6. *Get_user_input_data: the sign up function. It takes the input from the user (name, username password or generate for him a password if he wants) and return it to the main script for sav* 7. *User_login: allows the user to try to log into the system and adding a user which fails to type right password for 3 times to the blocked users file.* *__main__* *It will show the user an initial verification code from the random library with 6 characters, the user* *Will be asked to repeat this code, so he can proceed to the next step. If the user failed to match the password to its username for 3 times, the script will end.* *On the next step the user will be asked if he has an account of not. In case he has an account, the fu "user_login_ will be called, so he can log in.* *In case he does not have an account, the function "ger_user_input_data" will be called and the* *User will be able to sign up. In this step the script will try to read the users file if exists, and if it does exists, it will create the file with the following columns: name, username, password, strikes.* *In addition, it will save the return data from the sign up function, save it to the file and ask the new u he would like to log in.*

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3. How did you ensure that all parts of the script were covered?	
4. Write up external user-level documentation of how to use the project.	*The system will show you an initial code that you will need to repeat rightly, so you get verified.* *On the next screen you will be asked if you have an account or not. Be aware that you can try to rep the initial code rightly up to 3 times only.* *If "yes":* *You will access the log in system.* *In the log in system you will need to type in your existing username and your password.* *Be aware that if you don't type in your password correctly for 3 times in a raw, your account will get blocked and you won't be able to log in until you solve it with the administrator.* *If "no":* *You will access the sign up system.* *In the sing up system you will be asked to type you name in, to choose a user name and to be choos your own password (meeting the minimum requirements: 8 characters, lowercase letter, upper case and numbers) or to ask the system to generate a password for you. In case you use the password generator, it is very important to copy the password that it generated for you, and store in a safe plac you to remember.*

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5. Screenshot of comments in the script. Explaining variables, functions and interactions.	

```
# The log in system
2 usages
def user_login(users_data):
    # Checking if a file for blocked user exists in order to read it
    file_exists2 = exists("blocked_users.csv")
    if file_exists2:
        csv.DictReader("blocked_users.csv")
    else:
        # If the file does not exists, it will create a new file to store the blocked users
        with open('blocked_users.csv', 'w', newline='') as csvfile:
            fieldnames = [USER_NAME]
            writer = csv.DictWriter(csvfile, fieldnames=fieldnames)
            writer.writeheader()

    username = input("Please enter your username> ")
    while check_blocked_users(username):  # First it is checking the user is not blocked
        # If the user is blocked, the code will stop here
        print("\033[1;31;40m" "Your account is blocked. Please contact the administrator.")
        exit()
    while not check_user_login(username):  # If user is not blocked, it will check if it exists
        username= input(Fore.RED + "Invalid username, please try again> " + Fore.RESET)
```